

Tarvin Kaur Arora

Year 4, Statistics Major

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TECHNICAL SKILLS

Programming Languages: Java, Python, R, PHP

Databases: SQL, MongoDB

Data Science & Analysis: R (tidyverse, ggplot2, dplyr, glmnet), Python (pandas, NumPy, matplotlib, seaborn, scikit-learn), Excel

Software & Tools: PowerBI, Tableau, Git/Github, VS Code

OCR & Automation Frameworks: PyTesseract, pdf2image, FuzzyWuzzy

TECHNICAL PROJECTS

Data Pipeline and Database Design for Streaming Platform Analytics: Academic Project

January 2026

Technologies: SQL, MongoDB, Python, PyMongo, Pandas, Matplotlib

- Designed ER diagrams, relational schemas, and MongoDB schemas while enforcing key constraints, entity relationships, and data integrity.
- Developed SQL queries and MongoDB aggregation pipelines for data transformation, integration, and analytical reporting.
- Analyzed platform data to generate statistical insights to identify trends and evaluate engagement, pricing, and content performance.

Unsupervised Learning for Seismic Signal Characterization Using Self-Organizing Maps: Personal Project

June 2025

Technologies: Python, torch, MiniSom, Kmeans, scikit-learn

- Applied core signal-processing principles (spectral transforms, windowing, feature extraction) to interpret complex time-series data and computed fast-fourier transformations on data.
- Built and trained unsupervised learning models to discover latent structure and organize high-dimensional frequency patterns.
- Performed comparative analysis across multiple datasets using dimensionality-reduction techniques and similarity metrics.

Experimental Design and Statistical Analysis of Neural Network Preprocessing Strategies: Personal Project

September 2025

Technologies: Python, NumPy, Pandas, Torch, scikit-learn, ANOVA

- Designed and executed a computer simulated experiment to explore the impact of neural network model architecture (MLP, CNN, LSTM), preprocessing strategies, and capacity regimes on prediction performance.
- Applied core statistical theory through experiment design, while leveraging Latin square and fractional/full factorial designs to conduct ANOVA-based inference for accuracy and F1 score.
- Demonstrated strong statistical inference skills through interpretation of main effects, interaction effects, and validating ANOVA assumptions through residual diagnostic plots and Q-Q plots.

Applied Machine Learning for Credit Card Default Risk Prediction: Personal Project

October 2025

Technologies: Matplotlib, NumPy, Pandas, scikit-learn

- Engineered and preprocessed features after exploratory data-analysis (EDA) on the Credit Card Default dataset via imputation, scaling, and one-hot encoding.
- Trained and evaluated multiple models (tree-based ensembles and linear models), applying cross-validation and hyperparameter tuning to reduce overfitting and improve generalization.
- Conducted feature selection to identify important features driving default risk and interpreted performance while balancing model complexity and interpretability.

Textbook Rental Application: Academic Project

August 2024

Technologies: Java, Git, Junit Testing, VS Code, GitHub, OOD, Design Patterns

- Designed and implemented a user-friendly GUI through Object-Oriented Programming techniques, that allows users to rent and list textbooks on the marketplace.
- Executed incremental UI/UX development by integrating UML diagrams and app sketches to model class associations and tracking user interactions through JSON-based event logging.
- Integrated data persistence to fetch previous application states and the “Wishlist” option for users via JSON deserialization from JSON data.

Heart Disease Prediction with health predictors in Men: Academic Project

May 2023

Technologies: R, Excel, ggplot2, GitHub, glmnet

- Developed a predictive model in R, by leveraging machine learning techniques such as linear regression to predict heart disease risk in men aged 30-70 and achieved 81% accuracy through cross-validation and tuning.

- Investigated multiple univariate and bivariate analyses using appropriate graphical visualizations to explore variable relationships and correlations, supporting feature engineering and enhancing performance evaluation.

WORK EXPERIENCE

STEM and Language Tutor (IB French, Biology, Economics, AP Statistics and Math) - Varsity Tutors | *Remote, USA* **Current**

- Tutored students from grades 5 – 12 and college levels through personalized lesson plans, creative presentations, and strategy development tailored to unique academic requirements.
- Managed administrative tasks, and leveraged online platforms to create engaging lessons, simplifying complex topics through illustrations and analogies to make learning more accessible.
- Fulfilled instant tutoring requests and homework help questions, providing immediate answers across diverse subjects with minimal preparation before classes, maintaining 5-star rating record.

Metrics, Analytics & Reporting – Marsh McLennan (Mercer Consulting Private Limited) **June 2025 – August 2025**

- Built Darwin InfoSec Query Chat Assistant, reducing InfoSec Query response time (between client and legal teams) from 14 days to 2 minutes by automating OCR-Based Q&A Extraction (1,210 fields processed, 99% efficiency gain).
- Presented the Chat Assistant solution on a panel to Mercer's automation department for deployment, and training colleagues for replication across teams.
- Designed and deployed a PowerBI Total Remuneration Survey Dashboard with 12 KPIs, using PowerQuery, DAX, and dynamic visuals to deliver client benchmarking insights.
- Contributed to RAG Tracker automation, consulting on and AI system that generated Excel macros from pseudocode, improving project manager efficiency.
- Co-presented research findings on AI in Data Visualization, exploring predictive trends, persona-aware dashboards, and storytelling applications for decision-making.

Retail Operations Supervisor - BC Place Stadium | *Vancouver, BC* **July 2023 – December 2023**

- Supervised concession operations in a fast-paced environment, managing financial transactions and customer service for event crowds of 40,000 – 50,000 patrons.
- Trained and mentored a team of 6-8 employees on policies and procedures, ensuring efficient service under high-pressure conditions.
- Recommended equipment upgrades and inventory adjustments based on detailed sales report analysis and accurate logging of 100s of inventory items twice daily, completed within 20 minutes before and after events.

Barista – The Boulevard Coffee Roasting Co. | *Vancouver, BC* **August 2022 – August 2023**

- Navigated high-pressure, fast-paced service by processing customer orders daily with precision, and real-time problem solving.
- Strengthened communication, teamwork, and adaptability skills, coordinating with colleagues and engaging diverse customers to ensure positive and experience.
- Honed attention to detail and memory recall by mastering complex drink recipes, and multi-step processes under time pressure.

VOLUNTEER EXPERIENCE

Chief Executive Officer and Graphic Designer - Citizens Enlightened | *Remote, USA* **May 2019-February 2022**

- Approved and curated graphic designs for Instagram posts about global affairs and optimized social media using Search Engine Optimisation (SEO), and targeted content creation.
- Supervised content production, and ensured timeliness by managing scheduling, verifying broadcasted information, and effectively communicating expectations to the team.

Social Media Manager and Program Director - VCare Cancer | *Mumbai, India* **July 2020 – July 2022**

- Led a team of 20 volunteers in organizing the Breast Cancer Awareness Campaign, including interviewing survivors and doctors, creating educational content, and raising funds for patients through the Squash a Mile event.
- Coordinated volunteer schedules and campaign logistics, ensuring smooth collaboration across multiple teams and timely execution of events.

Crowdfunding and Outreach Campaigns - GreenSole Foundation & Helping Hands Foundation | *Remote* **August 2021**

- Raised ₹54100 (1500 CAD) to provide PPE kits to healthcare professionals in Mumbai during the COVID-19 pandemic, directly supporting frontline workers in crisis response.
- Raised ₹35,000 (560 CAD) to provide footwear for thousands of children in Maharashtra, India, who walk barefoot to school, improving their safety and comfort.

EDUCATION

University Of British Columbia

Expected Graduation: December 2026

- Bachelor of Science Major in Statistics